COSC265-26S2: Assignment 2

This assignment focuses on the use of the EER model to design a conceptual schema for a database for a veterinarian clinic. You should complete your work with EER-Tutor found at: https://ictg.csse.canterbury.ac.nz/eert/login.

This assignment is worth 16% of the overall grade for COSC265, and is due on Friday, 21 August 2026, at 5pm.

Submissions will be accepted for a week beyond this time, but a deduction of 15 marks will be made to your final grade. (So, if you got 80 of the 100 marks available, you would receive a final score of 65.) Submissions beyond this time will not be accepted.

You must prepare your final submission using EER-Tutor, using the free-hand mode (question 99). It is a good idea to develop your initial solution on paper. You must also generate a report. This is a simple document that should list the set of assumptions that you are making about the mini-world you are modelling. For each assumption briefly discuss: why you made the choice you did and any limitations that have been created as a result of the choice. Note that assumptions should not just be restatements of the requirements or be obvious consequences of the requirements. You should aim to limit your report to approximately 500 words — exceeding this limit should be done with care!

Submit your solution via AKO | LEARN. Your schema should be a .png image, and your report a .pdf document. Both should clearly display your full name and student ID.

Requirements

You are to develop an EER diagram that best reflects the requirements of a small veterinarian clinic.

Account

- Each user has their own account. The users name (first and last), postal address, email address, and contact number should be recorded.
- Each user has a unique username, and a password that they must provide to access their account.
- The date the account was created should be recorded. Accounts are either active or suspended. An admin user can set a future date for an account to be suspended.
- Each user is either a pet owner or a staff member.
- Pet owners must have at least one pet; they may subscribe to a weekly newsletter.

Staff

- Each staff member has a job title (Senior Vet, Receptionist, Technician, etc)
- The start date of a staff member is known.
- Staff members may be a system admin.
- Staff members are either on a fixed term or continuing contract
- If a staff member is on a fixed term contract, the end date of the contract is known.
- Staff members are either salaried or casual staff.
  The annual salary rate is stored for the former, the hourly rate for the latter.
- A staff member may be a Vet.
  - Vets have a unique certification number.
  - The date that a vet next needs to recertify is stored.

Pet

- A pet has a name (first, last: matches that of their owner); date of birth; species, breed, and current weight.
- A pet may have been microchipped, if they have then the country code and an identification number are stored.
- How old the pet is should be recorded.

Service

- Services are provided for a pet.
- A service is often one of three types:
    - Surgery
        - The type of surgery performed, and clinical notes should be recorded.
        - A surgery is conducted by a surgical team.
    - Checkup
        - A checkup is performed by a Vet.
        - The weight of the pet should be recorded.
    - Kennel stay

      the following aspects of a kennel stay are recorded:
        - The end date of the stay.
        - The type of kennel used.
        - Any dietary restrictions of the pet.
- The date as well as the cost of the service should be recorded.
- Post care instructions are stored. These notes outline any special requirements of care to be carried out.
- Extra notes about the service can also be stored.
- Each service is identified by a unique ticket number.

Surgical Team

- Each surgical team is composed of at least two members of staff.
- One of the team members is designated the team leader, this person must be a vet on a continuing contract.
- A member of staff may serve on more than one team.
- Only one team performs any given surgery service.
- Each team has a unique name.

Medicine

- Each medicine has a pharmacological name and a description of its common uses
- A medicine may interact with another medicine. One is called the primary agent, the other in referred to as the secondary agent. A description of the effects, and a severity score is recorded.
- A medicine of a particular type is ordered in a lot. A lot can be uniquely identified by the name of the pharmaceutical company that produced it, and a lot number.
- The date the order is placed, the date it is delivered to the clinic, and the expiry date of the medicine are recorded.
- The cost of the order, and the unit price that each dose will be sold at is recorded.
- Each dose of a medication is identified by a dose number, unique within the lot.
- A dose of a medicine may be administered to a pet during the course of a service.

Payment

- A payment is made for services rendered.
- The total cost to be paid is calculated based on the cost of individual services included.
- Each payment has a unique payment number.
- A discount may be applied at the discretion of the staff member.
- The payment method should be recorded.
- A copy of the receipt is sent to the owner (by email or post: at their discretion).

A few notes on marking

Your solution will be marked on several aspects:

- Appropriateness of constructs you use. In EER diagrams, often there are several ways of representing the same collection of data. However, some of these ways would be more appropriate than others, due to various factors, such as the semantics of the domain, the space required to store the data, the typical kinds of operations that will be performed on the database etc. Make sure you analyse the requirements properly.
- Make sure you use the correct EER notation and naming guidelines.
- You are allowed to introduce additional assumptions. However, not every assumption is reasonable. Think carefully about any assumption you would like to add. You are welcome to discuss your assumptions with the teaching team.